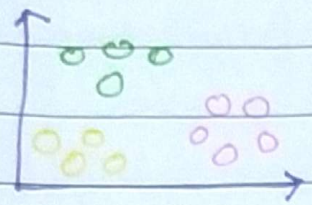
①

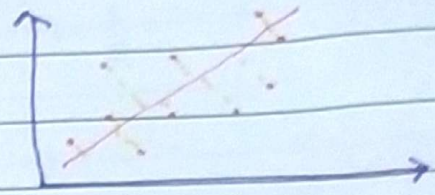
Types of unsupervised ML

- Clustring
- Dimensionality Reduction

** Put the simmilar data in the same Group



** Reduce the number of variables used



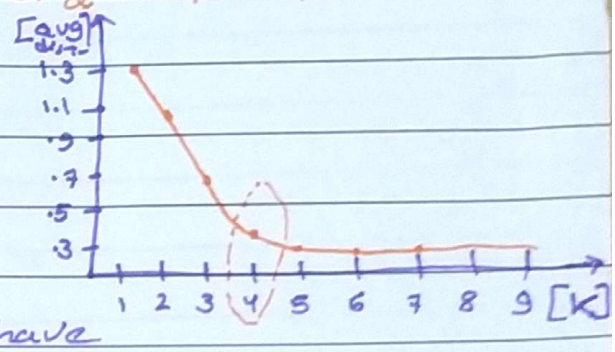
⇒ K means : used to cluster the data to many clusters

How to decide the value of k

- Visualization
- Pirior knolugo
- Discition methods

① Elbow Method

** Try different values of k and measure the average distance between the center of the cluster & the points.

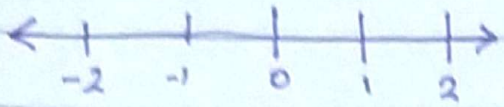


- The best choice of k is around the elbow of the curve
- Increasing k after the elbow doesn't have great effect on the average distance

CamScanner

feature scaling

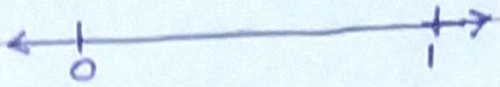
- Standrizing
- using z scores



→ Mean = 0 Variance = 1

→ used in clustring

- Normalizing
- Min max standrizi



→ Min = 0 max = 1

→ used in colouring

CamScanner